Compact S^3 Cosmology, Quasi-local Diffeomorphism Charges, and Discrete φ-Level Embedding

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Abstract

We develop a GR-compatible compact-topology framework for quantum cosmology on a spacetime $M = \mathbb{R} \times \Sigma$ with compact, orientable, boundaryless spatial hypersurfaces. The spatial sector is taken to be a spherical space form $\Sigma \simeq S^3/\Gamma$, with S^3 as the minimal simply connected representative. General covariance under $\mathrm{Diff}(M)$ implies Noether-II identities and first-class constraints rather than ordinary global Noether-I conserved charges. In a closed universe with $\partial\Sigma = \emptyset$, the global diffeomorphism Hamiltonian vanishes identically: the covariant phase-space charge is a boundary integral, and there is no boundary. Nontrivial physics is encoded in constraints, relational observables, and quasi-local boundary charges for subregions $D \subset \Sigma$ with $\partial D \neq \emptyset$. We provide an explicit Iyer–Wald derivation of the Noether charge two-form Q_ξ and the Gibbons–Hawking–York boundary three-form B, compute the Brown–York quasi-local energy for geodesic balls in closed FRW geometry, and compare it with the Misner–Sharp mass. We introduce a discrete φ-level embedding structure with smooth and piecewise-constant (floor) variants; the floor variant requires junction conditions at transition times to prevent distributional sources in the Euler–Lagrange equations. The appearance of the golden ratio φ follows from a uniqueness requirement on discrete scale decomposition (Zeckendorf–Lekkerkerker theorem), not from any empirical fit. Observational discriminants of compact topology arise through discrete Laplacian spectra in the CMB angular power spectrum and through infrared structure in the stochastic gravitational-wave background. We formulate explicit CMB and SGWB likelihood protocols based on closed-geometry Boltzmann solvers. The framework does not modify Einstein gravity and does not claim a detection; it provides a falsifiable compact-topology sector fully consistent with general relativity. We do not perform a numerical likelihood analysis in this work; instead, we provide a fully specified theoretical framework together with an explicit protocol for observational testing.

Keywords: compact topology, quantum cosmology, Noether identities, quasi-local charges, covariant phase space, gravitational waves

1 Introduction

Standard cosmological analyses commonly adopt spatially flat, non-compact $\mathbb{R}^3$ slices as a convenient background. This choice is well motivated by the near-zero spatial curvature

inferred from CMB power spectra and baryon acoustic oscillations [31]. It is, however, a choice: local near-flatness does not logically entail global non-compactness of the spatial topology. A sufficiently large sphere S^3 is locally indistinguishable from $\mathbb{R}^3$, just as the surface of the Earth appears flat on human scales. The question of global topology is therefore observationally open, even while curvature is tightly constrained.

Compact spatial topology is not merely a geometric curiosity. It changes the structure of the mode spectrum of perturbations (discrete versus continuous), alters the interpretation of conservation laws and Hamiltonian generators in GR, and potentially leaves imprints in CMB low-ℓ statistics and in the infrared spectrum of stochastic gravitational waves. These consequences make compact topology a falsifiable physical hypothesis, not an unfalsifiable metaphysical option.

In diffeomorphism-invariant gravity, the concept of "total energy of the Universe" requires careful reexamination. Diffeomorphisms are a local gauge redundancy parameterised by arbitrary vector fields $\xi^\mu(x)$. Noether's second theorem [1] therefore applies: gauge invariance generates differential identities among the Euler–Lagrange expressions (Bianchi-type relations) rather than independent global conserved charges of Noether-I type. In the covariant phase-space (CPS) formalism of Lee and Wald [3] and Iyer and Wald [5], the diffeomorphism Noether current is on-shell exact, $J_\xi \approx dQ_\xi$, so every charge is a boundary integral. For a compact, boundaryless slice $\partial\Sigma = \emptyset$, the boundary vanishes and with it the global Hamiltonian: $H_\xi[\Sigma] = 0$. This is not a failure of the theory but a deep structural feature of diffeomorphism-invariant theories on compact domains.

Nontrivial physical information survives in two forms: the constraint equations that must be satisfied on every Cauchy slice, and quasi-local boundary charges defined for subregions $D \subset \Sigma$ possessing a non-empty boundary $\partial D \neq \emptyset$. The quasi-local energy of geodesic balls in closed FRW geometry, computed via the Brown–York procedure [7], provides a concrete diagnostic of local physics even in the absence of a global energy.

A secondary aim of this paper is to introduce a discrete φ-level refinement structure as a bookkeeping device for cosmological scale indexing. The golden ratio $\varphi = (1+\sqrt{5})/2$ appears not as a phenomenological fit or an empirical coincidence, but as the unique asymptotic multiplier of the Fibonacci ladder, which is in turn selected by a uniqueness requirement on discrete scale decomposition (the Zeckendorf–Lekkerkerker theorem [40, 41]). The physical postulate is that this refinement ladder indexes cosmological scale structure; the mathematical content is theorem-level. Two implementations are provided: a smooth embedding that is fully compatible with the variational principle, and a piecewise-constant (floor) embedding that requires junction conditions at transition times.

Contributions

This paper makes the following contributions:

(i) A topology-explicit Noether-II formulation for compact spatial slices Σ with $\partial\Sigma = \emptyset$, establishing the vanishing of global diffeomorphism Hamiltonians as a corollary.

(ii) An explicit covariant phase-space derivation of the Iyer–Wald charge two-form Q_ξ and the Gibbons–Hawking–York boundary three-form B for Einstein–Hilbert gravity.

(iii) Computation of the Brown–York quasi-local energy for geodesic balls in closed S^3 FRW geometry, and its comparison with the Misner–Sharp mass.

(iv) A variationally consistent smooth/floor φ-level embedding with explicit junction conditions for the floor variant.

(v) CMB and SGWB observational discriminants and an explicit likelihood verification protocol.

The paper is organised as follows. Section 2 establishes the geometric and topological setup. Section 3 presents the Noether-II chain and the vanishing of global Hamiltonians. Section 4 gives explicit GR boundary charge formulas. Section 5 computes quasi-local energies in closed FRW. Section 6 discusses CMC slicing and York time. Section 7 introduces the φ-level embedding. Section 8 discusses observational discriminants and the likelihood protocol. Section 9 provides a discussion and Section 10 concludes.

2 Geometry and Topology of the Compact Sector

2.1 Spacetime decomposition

Definition 2.1 (Spacetime decomposition)**.** *Let $M = \mathbb{R} \times \Sigma$, where Σ is a compact, orientable 3-manifold without boundary, $\partial\Sigma = \emptyset$.*

We work in the 3+1 decomposition with metric

$$ds^2 = -(N^2 - N_i N^i)\, dt^2 + 2N_i\, dt\, dx^i + h_{ij}\, dx^i\, dx^j, \tag{1}$$

where N is the lapse, N^i the shift, and h_{ij} the induced metric on Σ_t. The Noether-II and topology statements derived below depend only on $\partial\Sigma = \emptyset$ and are insensitive to the detailed form of the metric.

2.2 Spherical sector

Diffeomorphism invariance does not by itself determine the global topology of Σ; topology is a sector of the gravitational configuration space. Two standard mathematical routes single out the spherical sector.

Theorem 2.1 (Hard spherical sector)**.** *Let Σ be a closed 3-manifold, i.e. compact and boundaryless, $\partial\Sigma = \emptyset$. If Σ is simply connected,*

$$\pi_1(\Sigma) = 0, \tag{2}$$

then Σ is homeomorphic to the 3-sphere: $\Sigma \cong S^3$.

This is the Poincaré–Perelman geometrisation criterion in dimension three [21, 22, 23, 24, 25].

Proposition 2.2 (Spherical space forms)**.** *Let Σ be closed, $\partial\Sigma = \emptyset$. If Σ admits a complete Riemannian metric of constant positive sectional curvature, then*

$$\Sigma \simeq S^3/\Gamma, \tag{3}$$

where $\Gamma \subset \mathrm{Isom}(S^3)$ is a finite group acting freely, and the universal cover is $\widetilde{\Sigma} \simeq S^3$ [18, 19].

If additionally Ric > 0, Hamilton's theorem [20] forces the space to be a spherical space form, providing a geometric (rather than purely topological) route to the same sector.

Remark 2.1. *We do not claim that observations uniquely identify* $\Gamma = \{1\}$ *(i.e.* $\Sigma \cong S^3$*). Present data constrain the injectivity radius of candidate fundamental domains and rule out certain identifications* Γ *on scales comparable to the last-scattering surface. The spherical sector* $\Sigma \simeq S^3/\Gamma$ *is treated as the relevant compact class, with* Γ *and the curvature radius* R_K *as free parameters to be constrained by likelihood analysis (Section 8).*

2.3 Mode spectra and the curvature radius

In a closed FRW slicing with curvature radius R_K, scalar harmonics on S^3 satisfy

$$\Delta_{S^3} Y_{n\ell m} = -\frac{n(n+2)}{R_K^2}\, Y_{n\ell m}, \quad n = 0, 1, 2, \ldots \tag{4}$$

We define the effective wavenumber $k_n = (n+1)/R_K$. The discreteness of this spectrum, with spacing $\Delta k = 1/R_K$, is the direct observational consequence of compact topology. The present-day curvature radius is

$$R_K^{(0)} = \frac{c}{H_0\sqrt{|\Omega_k|}}, \quad \Omega_k < 0, \tag{5}$$

and current Planck constraints require $R_K^{(0)}$ to be very large if $\Omega_k \approx 0$, so that discrete-mode signatures appear only at the lowest multipoles.

3 Noether II and the Vanishing of Global Diffeomorphism Hamiltonians

3.1 Central chain

The key result of this section may be summarised compactly:

$$\boxed{\text{Diff}(M) \Rightarrow \text{Noether II} \Rightarrow J_\xi \approx dQ_\xi \Rightarrow H_\xi = \int_{\partial\Sigma} \big(Q_\xi - \xi \cdot B\big) \Rightarrow \partial\Sigma = \emptyset \Rightarrow H_\xi[\Sigma] = 0.} \tag{6}$$

Each arrow in (6) is established below.

3.2 Noether-II identities

Theorem 3.1 (Noether II for diffeomorphism invariance)**.** *Let* $S[g, \psi]$ *be invariant under infinitesimal diffeomorphisms generated by an arbitrary smooth vector field* $\xi^\mu(x)$*:*

$$\delta_\xi g_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu, \qquad \delta_\xi \psi = \mathcal{L}_\xi \psi. \tag{7}$$

Then the Euler–Lagrange expressions satisfy differential identities (Noether-II identities):

$$\nabla_\mu E^{\mu\nu} + K^\nu(E_\psi, \nabla E_\psi, \ldots) \equiv 0, \tag{8}$$

where K^ν *is a covariant differential operator determined by the tensor character of* ψ*.*

Proof. Diffeomorphisms are parameterised by arbitrary functions $\xi^\mu(x)$, hence Noether's second theorem [1] applies. Varying S under (7) yields

$$0 = \delta_\xi S = \int_M d^4x\sqrt{-g}\,\big(E^{\mu\nu}\delta_\xi g_{\mu\nu} + E_\psi\,\delta_\xi\psi\big) + \int_M d^4x\sqrt{-g}\,\nabla_\mu(\cdots). \tag{9}$$

Integration by parts transfers derivatives from ξ^μ to $E^{\mu\nu}$ and E_ψ. Since $\xi^\mu(x)$ is arbitrary, the coefficient of ξ_μ must vanish identically, giving (8). □ □

Remark 3.1. *Equation* (8) *is an off-shell identity. It does not require the equations of motion to hold, and it is distinct from the Bianchi identity* $\nabla_\mu G^{\mu\nu} = 0$*, which follows as a consequence.*

3.3 Covariant phase-space structure

Treat the Lagrangian as a 4-form L and write its variation as

$$\delta L = E(\phi)\,\delta\phi + d\Theta(\phi, \delta\phi), \tag{10}$$

where $E(\phi) = 0$ are the field equations and Θ is the symplectic potential 3-form. For a vector field ξ generating a diffeomorphism, define the Noether current 3-form

$$J_\xi := \Theta(\phi, \mathcal{L}_\xi\phi) - \xi \cdot L. \tag{11}$$

A standard identity gives $dJ_\xi = -E(\phi)\,\mathcal{L}_\xi\phi$, so $dJ_\xi \approx 0$. For gauge symmetries such as diffeomorphisms, Noether II strengthens this to on-shell exactness:

$$J_\xi \approx dQ_\xi, \tag{12}$$

for some locally constructed 2-form Q_ξ (the Noether charge form). Equation (12) is the clean statement that covariant charges in GR are generically boundary charges [3, 5].

3.4 Hamiltonian generators and their compact limit

The variation of the Hamiltonian generator associated with ξ over a spatial region $D \subset \Sigma$ is

$$\delta H_\xi[D] = \int_{\partial D} \big(\delta Q_\xi - \xi \cdot \Theta(\phi; \delta\phi)\big). \tag{13}$$

When boundary conditions imply integrability,

$$\xi \cdot \Theta(\phi; \delta\phi) = \delta(\xi \cdot B) \quad \text{on the chosen phase space,} \tag{14}$$

one obtains an actual Hamiltonian generator (up to an additive constant):

$$H_\xi[D] = \int_{\partial D} \big(Q_\xi - \xi \cdot B\big). \tag{15}$$

The B-term arises from ambiguities in Θ (defined up to dB) and ensures integrability on phase space; the Gibbons–Hawking–York choice of B is made explicit in Section 4.

Corollary 3.2 (Compact boundaryless Σ)**.** *If Σ is compact and boundaryless, $\partial\Sigma = \emptyset$, then for any smooth gauge diffeomorphism parameter ξ,*

$$H_\xi[\Sigma] = \int_{\partial\Sigma} \big(Q_\xi - \xi \cdot B\big) = 0. \tag{16}$$

In particular, for $\Sigma \cong S^3$ or $\Sigma \simeq S^3/\Gamma$, there is no global Noether-I "energy of the Universe".

Remark 3.2 (Physical interpretation)**.** *The vanishing* (16) *is not a pathology. It is the statement that, on a compact boundaryless slice, the Hamiltonian generator of gauge diffeomorphisms is a sum of first-class constraints that vanish on physical states. Physical information is carried by three structures: (i) the constraint equations* (8) *that must hold on every Cauchy slice; (ii) relational observables constructed from correlations among fields; and (iii) quasi-local boundary charges $H_\xi[D]$ for subregions $D \subset \Sigma$ with $\partial D \neq \emptyset$. The absence of a global bulk energy is, in fact, the gravitational avatar of the fact that the gravitational stress-energy is a pseudotensor: no local, gauge-invariant energy density exists in GR [4].*

Remark 3.3 (Explicit distinction: Σ versus D)**.** *Throughout this paper two distinct domains appear, and the distinction is central to the physics.*

- *The* full Cauchy slice Σ *is compact and boundaryless: $\partial\Sigma = \emptyset$. For any smooth gauge diffeomorphism parameter ξ, the global Hamiltonian generator vanishes: $H_\xi[\Sigma] = 0$.*
- *A* subregion $D \subset \Sigma$ *is a proper subset with non-empty boundary: $\partial D \neq \emptyset$. The quasi-local generator $H_\xi[D] = \int_{\partial D}(Q_\xi - \xi \cdot B)$ is generically non-zero and carries physically meaningful information.*

The vanishing of $H_\xi[\Sigma]$ therefore does not imply that $H_\xi[D] = 0$; it implies only that the global generator has no boundary support. Local physics is fully captured by $H_\xi[D]$ for appropriately chosen subregions D.

3.5 ADM canonical language

In the ADM decomposition, $\xi^\mu = N n^\mu + N^\mu_\parallel$ with lapse N and shift N^i. The canonical generator is

$$H_{\rm ADM}[N, N^i] = \int_\Sigma (N\mathcal{H} + N^i \mathcal{H}_i) + \oint_{\partial\Sigma} B[N, N^i], \tag{17}$$

where $\mathcal{H} \approx 0$ and $\mathcal{H}_i \approx 0$ are the Hamiltonian and momentum constraints, and B is the GHY boundary functional needed for differentiability. On-shell the bulk vanishes, and for $\partial\Sigma = \emptyset$ the boundary term vanishes as well: $H_{\rm ADM} = 0$ on-shell, consistently with Corollary 3.2.

4 Explicit GR Boundary Charges

4.1 Einstein–Hilbert Lagrangian

We specialise to GR with optional cosmological constant Λ. The Einstein–Hilbert Lagrangian 4-form is

$$L_{\rm EH} = \frac{1}{16\pi G}(R - 2\Lambda)\,\epsilon, \tag{18}$$

where ϵ is the spacetime volume 4-form. The variation (10) yields

$$\Theta_{abc}(g;\delta g) = \frac{1}{16\pi G}\,\epsilon_{dabc}\,(\nabla^e \delta g^d{}_e - \nabla^d \delta g), \tag{19}$$

where $\delta g := g^{mn}\delta g_{mn}$.

4.2 Iyer–Wald Noether charge

The Iyer–Wald Noether charge 2-form is [5]

$$(Q_\xi)_{ab} = -\frac{1}{16\pi G}\,\epsilon_{abcd}\,\nabla^c \xi^d. \tag{20}$$

This reduces to the Komar integrand when ξ is a Killing vector: $Q_\xi = -(1/16\pi G)\epsilon_{\mu\nu}\nabla^\mu \xi^\nu$. For a general (non-Killing) diffeomorphism, the same formula holds but the associated quasi-local generator need not be conserved under time evolution.

4.3 Gibbons–Hawking–York boundary term

To make the variational principle well-posed on a manifold with a spacetime boundary hypersurface $\mathcal{B}$ under Dirichlet boundary conditions (fixed induced metric on $\mathcal{B}$), one adds the Gibbons–Hawking–York (GHY) term [10, 11]:

$$S_{\mathrm{GHY}} = \frac{1}{8\pi G}\int_{\mathcal{B}} \sqrt{|h|}\,K\,d^3x, \tag{21}$$

corresponding to the boundary 3-form

$$B_{\mathrm{GHY}} = \frac{1}{8\pi G}\,K\,\epsilon_{\mathcal{B}}, \tag{22}$$

where K is the trace of the extrinsic curvature of the 3-dimensional boundary $\mathcal{B} \subset M$ in spacetime, and $\epsilon_{\mathcal{B}}$ is its induced volume form. In quasi-local Hamiltonian constructions, boundary functionals are ultimately evaluated on 2-surfaces $\partial D \subset \Sigma$, requiring an additional reference subtraction (or counterterm) to fix the energy zero-point.

4.4 Relation to asymptotic charges

When $\partial\Sigma \neq \emptyset$ and the boundary is taken to infinity in an asymptotically flat spacetime, the formula (15) reproduces the ADM mass and angular momentum, and at null infinity it reproduces BMS charges. A closed universe $\Sigma \simeq S^3/\Gamma$ has no such asymptotic boundary: there is no analogue of the ADM mass defined as a global integral. This is physically appropriate, since the compact universe has no external observer at infinity. Meaningful energy diagnostics are available only quasi-locally, as in the next section.

4.5 Covariant phase-space ambiguities

The standard ambiguities of the formalism – $L \to L + d\mu$, $\Theta \to \Theta + \delta\mu + d\nu$ – induce shifts in (J_ξ, Q_ξ) and in the Hamiltonian by boundary terms. The choice of B (including the GHY term and a reference subtraction) fixes both integrability and the zero-point of quasi-local energy. A detailed treatment of these ambiguities is provided in Appendix A.

5 Closed FRW on S^3 and Quasi-local Energy

5.1 Geometry of the closed FRW slice

Consider a closed FRW slicing with spatial metric on $\Sigma_t \simeq S^3$:

$$d\ell^2 = a(t)^2\big[d\chi^2 + \sin^2\chi\, d\Omega_2^2\big], \quad \chi \in [0,\pi]. \tag{23}$$

Let $D(\chi_0)$ be a geodesic ball $0 \le \chi \le \chi_0$ with boundary $\partial D \simeq S^2$ at $\chi = \chi_0$. The areal radius and area are

$$R(\chi_0) = a(t)\sin\chi_0, \quad A(\chi_0) = 4\pi R^2, \tag{24}$$

and the volume is

$$V(\chi_0) = 2\pi a(t)^3\big(\chi_0 - \sin\chi_0\cos\chi_0\big). \tag{25}$$

5.2 Brown–York quasi-local energy

The Brown–York energy of D is [7]

$$E_{\rm BY} = \frac{1}{8\pi G}\int_{\partial D}(k_0 - k)\, dS, \tag{26}$$

where k is the trace of the extrinsic curvature of ∂D as embedded in Σ_t (with outward unit normal in Σ_t), and k_0 is a reference term fixing the energy zero-point. For $\chi = \chi_0$ spheres in $\Sigma \simeq S^3$, the extrinsic curvature evaluates to (derivation in Appendix B)

$$k(\chi_0) = \frac{2}{a}\cot\chi_0 = \frac{2}{R}\sqrt{1 - \frac{R^2}{a^2}}, \qquad k_0 = \frac{2}{R}, \tag{27}$$

where the reference $k_0 = 2/R$ is the mean curvature of an isometric embedding of the same S^2 intrinsic metric into $\mathbb{R}^3$. Substituting (27) into (26) yields the closed expression

$$E_{\rm BY}(R,a) = \frac{R}{G}\left(1 - \sqrt{1 - \frac{R^2}{a^2}}\right), \quad 0 \le R \le a(t). \tag{28}$$

For small balls $R \ll a$,

$$E_{\rm BY}(R,a) = \frac{R^3}{2Ga^2} + \mathcal{O}\left(\frac{R^5}{a^4}\right). \tag{29}$$

At the equator $R \to a$, $E_{\rm BY} \to R/G$. The non-trivial quasi-local energy (28) reflects the positive spatial curvature of S^3 relative to the Euclidean reference, and it is present even before any dynamics is invoked. This result was previously obtained by Afshar [15] in an equivalent coordinate system.

5.3 Misner–Sharp mass and comparison

In spherically symmetric spacetimes, the Misner–Sharp (MS) mass provides a spacetime-covariant quasi-local energy [8], often associated with the Kodama vector field [16, 17]:

$$M_{\rm MS}(R) = \frac{R}{2G}\big(1 - g^{ab}\nabla_a R\nabla_b R\big). \tag{30}$$

For closed FRW with $k = +1$,

$$M_{\rm MS}(R) = \frac{R^3}{2G}\left(H^2 + \frac{1}{a^2}\right), \quad H := \dot{a}/a. \tag{31}$$

Using the Friedmann constraint,

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \tag{32}$$

the Misner–Sharp mass for $k = +1$ evaluates to

$$M_{\rm MS}(R) = \frac{4\pi}{3}R^3\rho + \frac{\Lambda R^3}{6G}. \tag{33}$$

Equations (28) and (31) are complementary rather than identical. $E_{\rm BY}$ depends on the intrinsic and extrinsic geometry of ∂D as embedded in the chosen spatial slice Σ_t, plus the chosen reference subtraction; it is therefore tied to a slicing and boundary conditions. $M_{\rm MS}$ depends on $\nabla_a R$ and thus on the full spacetime geometry including the expansion H; it is covariant in spherical symmetry and directly relates to matter content through (33). A precise identification between $E_{\rm BY}$ and $M_{\rm MS}$ requires additional dynamical input and/or a preferred evolution vector field, and is not automatic from kinematical definitions alone [9].

6 Relational Time: CMC Slicing and York Time

In a compact universe, the global Hamiltonian constraint $\mathcal{H} \approx 0$ (Corollary 3.2) implies that dynamics must be described relationally rather than with reference to an absolute external time. A natural canonical choice for the relational clock is constant-mean-curvature (CMC) slicing [12, 13].

Let h_{ij} be the induced metric on Σ_t and K_{ij} its extrinsic curvature in spacetime; define the mean curvature

$$K := h^{ij}K_{ij}. \tag{34}$$

CMC slicing imposes $K = \text{const}$ on each Σ_t. For standard FRW slicings, K is spatially constant automatically by homogeneity and isotropy, so the usual cosmological foliation is already a CMC foliation. York time is then defined by

$$T_Y := -\frac{1}{3}K, \tag{35}$$

which increases monotonically during expansion and provides a canonical relational time variable compatible with constraint dynamics [14].

In the discrete φ-level framework introduced in the next section, the level index n (or $N(t)$) may be treated as a monotonic function of T_Y:

$$n = n(T_Y), \tag{36}$$

providing a relational clock compatible with the vanishing of global gauge generators on the closed slice. We emphasise that (36) is a modelling assumption, not a dynamical theorem.

7 Discrete φ-Level Embedding and Variational Consistency

7.1 Motivation: the Zeckendorf–Fibonacci uniqueness

The discrete φ-level embedding rests on two structural postulates and a spectral-geometric consistency condition, which we state explicitly before deriving the role of φ.

Postulate (H2) [Refinement uniqueness]. *The cosmological scale index is required to be path-independent: successive coarse-graining steps must compose associatively and yield a unique, unambiguous representation of each scale level regardless of the order of refinement.*

Postulate (H3) [Spectral-geometric consistency]. *The asymptotic multiplier of the scale ladder must coincide with the dominant eigenvalue of the minimal positive second-order linear recursion consistent with (H2).*

The sharp mathematical content of (H2) is the Zeckendorf–Lekkerkerker property [40, 41]: uniqueness of representation in the Fibonacci basis. The content of (H3) is then fixed by the Perron–Frobenius theorem applied to the companion matrix of the Fibonacci recursion $F_{n+2} = F_{n+1} + F_n$, whose companion matrix is $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ with dominant eigenvalue $\varphi = (1 + \sqrt{5})/2$.

Remark 7.1. *The golden ratio φ is* not *introduced as a free parameter. It arises as the dominant eigenvalue of the minimal positive second-order recursion via the Perron–Frobenius theorem. Its appearance in the cosmological framework follows from the structural postulates (H2) and the spectral-geometric consistency condition (H3), rather than from Einstein's equations alone.*

For a bookkeeping of cosmological scale structure to be path-independent in the sense of (H2), the representation of integers in the chosen refinement basis must be unique. The sharp mathematical form of this requirement is the Zeckendorf–Lekkerkerker property [40, 41].

Lemma 7.1 (Zeckendorf uniqueness and the golden ratio)**.** *Let $\{F_n\}$ be the Fibonacci sequence defined by $F_{n+2} = F_{n+1} + F_n$ with $F_0 = F_1 = 1$. Every positive integer admits a unique decomposition into a sum of non-adjacent Fibonacci numbers (Zeckendorf's theorem [40, 41]). The ratio $r_n = F_{n+1}/F_n$ converges to the unique positive root of $r^2 = r + 1$:*

$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n} = \varphi \equiv \frac{1+\sqrt{5}}{2}. \tag{37}$$

A weak converse [42] shows that this uniqueness requirement essentially characterises the Fibonacci ladder among monotone sequences. The associated asymptotic scale multiplier is therefore fixed as φ by the uniqueness requirement alone, up to initial normalisation.

Remark 7.2 (Anti-numerology statement)**.** *The appearance of φ in the present construction is not a phenomenological fit and not post-hoc pattern matching. It follows from the adopted refinement principle: uniqueness and path-independence of discrete scale decomposition. The empirical content of the framework is not that "φ appears in data", but rather that a compact-topology model with this refinement structure predicts specific*

discrete-mode or log-periodic features that can be tested against observational likelihoods. Whether these features are observationally significant is an empirical question left to the likelihood protocol of Section 8.

7.2 Embedding variables

Define the smooth embedding variable

$$\nu(t) = \log_\varphi\left(\frac{ct}{\ell_P}\right), \tag{38}$$

and the discrete floor embedding level

$$N(t) = \lfloor \nu(t) \rfloor \in \mathbb{Z}. \tag{39}$$

Two variational formulations are available:

(I) *Smooth embedding*: $S = \int d^4x\sqrt{-g}\,\mathcal{L}(\varphi, \nabla\varphi; \nu(t))$. This is differentiable and the variational principle is standard.

(II) *Floor embedding*: $S = \int d^4x\sqrt{-g}\,\mathcal{L}(\varphi, \nabla\varphi; N(t))$. $N(t)$ is piecewise constant, creating a sequence of transition times $\{t_k\}$ at which it increments. Junction conditions are required.

7.3 Junction conditions for the floor embedding

Splitting the floor-embedding action into intervals $I_k = (t_k, t_{k+1})$ and integrating by parts yields

$$\delta S = \sum_k \int_{I_k} d^4x\sqrt{-g}\,E_\phi\,\delta\phi + \sum_k \int_{\Sigma_{t_k}} d^3x\,[\Pi_\phi\,\delta\phi]_{t_k^-}^{t_k^+}, \tag{40}$$

where $\Pi_\phi \equiv \partial(\sqrt{-g}\mathcal{L})/\partial(\partial_t\phi)$ is the canonical momentum.

Theorem 7.2 (Junction consistency)**.** *In the floor-embedding formulation (II), the Euler–Lagrange equations are distribution-free if and only if*

$$\int_{\Sigma_{t_k}} d^3x\,\left[\Pi_\phi\,\delta\phi\right]_{t_k^-}^{t_k^+} = 0 \quad \textit{for all } k \textit{ and all admissible } \delta\phi. \tag{41}$$

A sufficient condition is continuity of canonical data at each transition time:

$$\phi(t_k^-) = \phi(t_k^+), \qquad \Pi_\phi(t_k^-) = \Pi_\phi(t_k^+). \tag{42}$$

Proof. Equation (40) is exact from the variational principle. Stationarity requires both the bulk Euler–Lagrange terms and the junction contributions to vanish for all admissible variations, giving (41). The conditions (42) imply that the junction integrands vanish pointwise, which is sufficient. □ □

7.4 Covariance and Noether-II consistency

The underlying gravitational sector remains diffeomorphism-invariant in both formulations. In the smooth formulation, $\nu(t)$ may be treated as an effective scalar embedding variable, and the Noether-II identity structure (Theorem 3.1) holds unmodified. In the floor formulation, $N(t)$ is piecewise-constant with a chosen foliation; adding the junction terms and imposing (42) ensures that the Euler–Lagrange system is distribution-free on each interval I_k, so Noether-II identities hold on each I_k and are globally consistent across transition hypersurfaces.

Remark 7.3 (Prototype matter sector)**.** *As a minimal prototype, one may consider a scalar field ϕ with φ-dependent potential $V(|\phi|; N) = \frac{1}{2}m^2|\phi|^2\varphi^{-2N}$, where ϕ is the matter field and φ is the golden ratio. This is illustrative only; the core GR results of Sections 3–5 are independent of any specific matter model. Model-dependent overlays are recorded in Appendix E.*

8 Compact Mode Spectra and Observational Discriminants

Scope of this section. This section formulates observational discriminants and a verification protocol. No numerical parameter estimation or likelihood evaluation is performed in the present work. The framework is presented as a falsifiable theoretical prediction map whose quantitative evaluation is deferred to future work.

8.1 Discrete CMB angular power spectrum

For a compact slice $\Sigma \simeq S^3/\Gamma$, the primordial scalar spectrum is a discrete sum rather than a continuous integral. Using the eigenvalue relation (4) with wavenumber $k_n = (n+1)/R_K$, the CMB angular power spectrum becomes

$$C_\ell = 4\pi \sum_{n=0}^{\infty} \mathcal{P}_{\mathcal{R}}(k_n)|\Delta_\ell(k_n)|^2, \tag{43}$$

where $\mathcal{P}_{\mathcal{R}}(k_n)$ is the primordial curvature power spectrum evaluated at k_n, and $\Delta_\ell(k_n)$ are the radiative transfer functions computed in closed FRW geometry. The fundamental angular scale is

$$\ell_{\min} \sim \frac{\chi_*}{R_K}, \tag{44}$$

where χ_* is the comoving distance to the last scattering surface. For R_K very large, the mode spacing $\Delta k = 1/R_K$ is small and the discrete sum approaches the standard continuous integral, recovering the flat-topology limit. Compact-topology effects are therefore most visible at low ℓ and are controlled by the single parameter R_K.

Remark 8.1 (Toy numerical example)**.** *For an illustrative curvature radius $R_K = 50\,\mathrm{Gpc}$, the first several scalar modes have $k_n = (n+1)/R_K$ with spacing $\Delta k = 0.02\,\mathrm{Gpc}^{-1}$ and degeneracy $(n+1)^2$. Table 1 lists these for $n = 0, \ldots, 10$. This is a minimal, checkable numerical instance of the replacement $k \to k_n$ in* (43)*.*

Table 1: Toy S^3 scalar modes for $R_K = 50\,\mathrm{Gpc}$. Here $k_n = (n+1)/R_K$ and the scalar degeneracy is $(n+1)^2$.

n	k_n [Gpc^{-1}]	Δk [Gpc^{-1}]	Degeneracy $(n+1)^2$
0	0.02	0.02	1
1	0.04	0.02	4
2	0.06	0.02	9
3	0.08	0.02	16
4	0.10	0.02	25
5	0.12	0.02	36
6	0.14	0.02	49
7	0.16	0.02	64
8	0.18	0.02	81
9	0.20	0.02	100
10	0.22	0.02	121

8.2 Matched circles for $\Sigma \simeq S^3/\Gamma$

For non-trivial holonomy group $\Gamma \neq \{1\}$, the spatial identification introduces matched-circle correlations in the CMB temperature and polarisation maps [26]. The absence of a statistically significant signal in current data places lower bounds on the injectivity radius of the fundamental domain [27, 28, 29]. Within the spherical sector, the group label Γ and the curvature radius R_K constitute the topological parameters to be constrained.

8.3 Stochastic gravitational-wave background

Tensor harmonics on S^3 are also discrete. The eigenvalue relation for symmetric traceless transverse tensor modes gives

$$\Delta_{S^3} T_{ij}^{(n)} = -\frac{\lambda_n^{(T)}}{R_K^2}\, T_{ij}^{(n)}, \quad n \geq 2, \tag{45}$$

with effective wavenumber $k_n^{(T)} \sim \sqrt{\lambda_n^{(T)}}/R_K$ and corresponding physical frequency $f_n = c\, k_n^{(T)}/(2\pi a_0)$. The gravitational-wave energy density spectrum is schematically

$$\Omega_{\mathrm{GW}}(f) \sim \sum_{n\geq 2} \mathcal{P}_T(k_n^{(T)})\, \mathcal{T}_T^2(k_n^{(T)})\, \delta(f - f_n), \tag{46}$$

implying an IR cutoff and possible step or modulated features set by the scale $c/(a_0 R_K)$.

Equation (46) is schematic. In any realistic analysis the δ-comb is broadened by finite observation time, redshift smearing, and the bandwidth of estimators used in PTA and LVK searches. For curvature radii compatible with current near-flat constraints, the mode spacing in PTA/LVK bands is $\Delta f/f \sim 1/n$ with $n \sim 2\pi f a_0 R_K/c \gg 1$, so the discrete structure manifests as weak modulations on a smooth background rather than sharp spectral lines. Current PTA detections are well described by a smooth power-law spectrum consistent with astrophysical SMBHB populations [38, 39]; therefore topology-induced modulations must lie in a marginally allowed regime or require next-generation sensitivity.

8.4 Likelihood and verification protocol

A credible submission requires that the model be implemented as a prediction map $\theta \mapsto \{C_\ell, \Omega_{\rm GW}(f), \ldots\}$ and tested against public likelihoods. We specify the minimal protocol below.

CMB

Let $\mathbf{D}$ denote the binned Planck data vector [30, 31] (e.g. TT, TE, EE) and $\mathbf{M}(\theta)$ the model prediction from a closed-geometry Boltzmann solver (CLASS or CAMB) [32, 33], with the replacement $k \to k_n = (n+1)/R_K$ in the mode sums (43). The Gaussian-approximate log-likelihood reads

$$-2\ln\mathcal{L}_{\rm CMB}(\theta) = \big(\mathbf{D}-\mathbf{M}(\theta)\big)^{\top}\mathbf{C}^{-1}\big(\mathbf{D}-\mathbf{M}(\theta)\big), \tag{47}$$

with covariance $\mathbf{C}$ provided by the likelihood package. Model selection uses $\Delta\chi^2 = -2\Delta\ln\mathcal{L}$ against the ΛCDM baseline at matched nuisance treatment, or a Bayesian evidence ratio

$$B_{\Gamma_1\Gamma_2} = \frac{Z_{\Gamma_1}}{Z_{\Gamma_2}}, \qquad Z_\Gamma = \int d\theta\, \mathcal{L}(\mathbf{D}|\theta,\Gamma)\,\pi(\theta|\Gamma), \tag{48}$$

within the spherical sector, where $\pi(\theta|\Gamma)$ is the prior.

SGWB

Given a prediction $\Omega_{\rm GW}(f;\theta)$, compare to LVK [37] and PTA constraints via their published likelihoods. A simple summary statistic is

$$\chi^2_{\rm GW}(\theta) = \sum_i \frac{\big(\Omega_{\rm GW}(f_i;\theta) - \Omega_i^{\rm obs}\big)^2}{\sigma_i^2}, \tag{49}$$

for an appropriate set of frequency bins $(f_i, \Omega_i^{\rm obs}, \sigma_i)$.

Non-negotiable deliverables

A complete future submission must include: (i) a reproducible code path generating C_ℓ and $\Omega_{\rm GW}(f)$ from the equations of this manuscript; (ii) a parameter table with priors; and (iii) a $\Delta\chi^2$ or Bayes factor report against standard baselines.

Caveat. We do not claim a detection or a statistical preference for compact topology in this manuscript. The present work supplies the GR-consistent theoretical framework and the explicit observational pipeline. Numerical likelihood evaluation is deferred to future work.

9 Discussion

Consistency if future likelihoods show no deviation. If future likelihood analyses show no statistically significant deviation from ΛCDM within the compact-topology sector, the present framework remains internally consistent but observationally unconstrained. The absence of a detection would narrow the viable range of (R_K, Γ) without falsifying the theoretical structure: it would exclude specific regions of parameter space

while leaving the GR-compatible compact-topology sector intact at larger curvature radii. A definitive exclusion of the entire compact spherical sector would require sensitivity to scales $R_K \lesssim R_{\rm LSS}$, which is beyond the reach of current experiments.

Local flatness versus global topology. A very large S^3 looks locally flat, just as the Earth's surface appears flat on human scales. The observational constraints $|\Omega_k| \lesssim 0.01$ from CMB+BAO analyses require $R_K \gtrsim 10\, c/H_0 \sim 440\,\mathrm{Gpc}$ if the universe is negatively curved, or correspondingly large for the spherical case. At such radii, the discrete mode spacing $\Delta k = 1/R_K$ falls below the resolution of current CMB experiments for all but the lowest multipoles. The compact-topology hypothesis therefore resides in a corner of parameter space that is allowed but not yet decisively tested. Future CMB polarisation surveys and PTA arrays with extended baselines may reach the required sensitivity.

Global Hamiltonian vanishing is not a pathology. The result $H_\xi[\Sigma] = 0$ (Corollary 3.2) is sometimes misread as implying that physics is trivial in a compact universe. This is incorrect. The constraints $\mathcal{H} \approx 0$ and $\mathcal{H}_i \approx 0$ encode non-trivial dynamics through the Friedmann equations and the evolution of perturbations. The quasi-local charges $H_\xi[D]$ for subregions $D \subset \Sigma$ carry physically meaningful information about local energy content, as demonstrated by the Brown–York computation in Section 5. The problem of time in quantum gravity on compact slices is resolved relationally, for example via York time (35).

Quasi-local charges retain physical content. The Brown–York energy (28) and the Misner–Sharp mass (31) are complementary quasi-local diagnostics. $E_{\rm BY}$ is naturally adapted to a chosen slicing and is directly measurable in principle by an observer at ∂D. $M_{\rm MS}$ is covariant in spherical symmetry and ties the local energy to matter content through the Friedmann constraint. Neither requires a global boundary; both are well-defined in the compact S^3 geometry.

Compatibility of φ-embedding with Noether-II. The smooth φ-embedding (formulation I) introduces $\nu(t)$ as an effective scalar embedding variable; the variational principle is standard and Noether-II identities hold without modification. The floor embedding (formulation II) requires junction conditions (42) at transition times $\{t_k\}$ to prevent distributional sources in the Euler–Lagrange equations for the matter field ϕ; once these conditions are imposed, the Euler–Lagrange system is distribution-free and Noether-II identities hold on each interval. The φ-level structure is therefore variationally consistent with diffeomorphism invariance in both formulations.

What would falsify the framework. The framework makes the following falsifiable predictions:

(a) The CMB angular power spectrum (43) deviates from the flat-topology prediction at low ℓ through a discrete-mode cutoff and modulation parameterised by R_K and Γ.

(b) Matched-circle correlations in the CMB sky map appear for specific angular scales determined by the fundamental domain of Γ.

(c) The SGWB shows an IR cutoff and weak modulations at frequencies set by $c/(a_0 R_K)$.

Non-detection of any of these features at sensitivity levels compatible with the predicted signal would rule out the corresponding region of (R_K, Γ) parameter space. A definitive exclusion of the entire compact spherical sector requires constraining R_K to values comparable to or smaller than the last-scattering radius, which current and near-future data are not yet able to do.

10 Conclusion

We have developed a compact spherical cosmology framework that is fully compatible with general relativity and capable of falsifiable observational predictions. The main results are the following.

The vanishing of global diffeomorphism Hamiltonians on compact boundaryless slices (Corollary 3.2) follows directly from the Noether-II structure of GR: the Noether charge is a boundary integral, and there is no boundary when $\partial\Sigma = \emptyset$. This is not a deficiency but the correct physical description of a closed universe. Nontrivial physics resides in the constraint equations and in quasi-local boundary charges for subregions.

Quasi-local charges provide meaningful diagnostics even in the absence of a global energy. The Brown–York energy (28) of geodesic balls in closed S^3 FRW geometry is computed in closed form, and its relationship to the Misner–Sharp mass and the Friedmann constraint is clarified.

The discrete φ-level embedding can be made variationally consistent. The smooth formulation requires no modification of the standard variational principle. The floor formulation requires junction conditions at transition times, which impose continuity of canonical data; once satisfied, the Noether-II structure is preserved. The golden ratio φ enters as the unique asymptotic multiplier of the Fibonacci ladder, which is selected by uniqueness and path-independence of discrete scale decomposition.

Observational tests are available through: (i) the CMB angular power spectrum at low ℓ, controlled by R_K and Γ; (ii) matched-circle statistics in the CMB for non-trivial holonomy; and (iii) infrared features and modulations in the stochastic gravitational-wave background. These predictions can be evaluated against Planck likelihoods and PTA/LVK constraints using closed-geometry Boltzmann codes and standard Bayesian model comparison.

Future work must implement the numerical likelihood pipeline: closed-geometry mode functions in CLASS/CAMB, MCMC/nested-sampling posterior analysis, and $\Delta\chi^2$/Bayes-factor reports against ΛCDM. Only after such an analysis can a statistical preference (or exclusion) for the compact spherical sector be claimed.

A Covariant Phase-Space Ambiguities

The covariant phase-space formalism contains standard ambiguities. The Lagrangian is defined up to an exact form: $L \to L + d\mu$. Under this shift,

$$\Theta \to \Theta + \delta\mu + d\nu, \tag{50}$$

for some 2-form ν. Correspondingly, the Noether current shifts as $J_\xi \to J_\xi + \mathcal{L}_\xi\mu$ and the charge form shifts as

$$Q_\xi \to Q_\xi + \xi \cdot \mu + d(\cdots). \tag{51}$$

The choice of boundary counterterm B (including the GHY term) is designed to cancel these ambiguities for a given class of boundary conditions and to fix the zero-point of quasi-local energy. In particular, the Euclidean reference $k_0 = 2/R$ used in Section 5 is one specific choice; other reference subtractions are possible and would shift E_{BY} by an R-dependent constant. The physical content is in differences $E_{\mathrm{BY}}(R_2, a) - E_{\mathrm{BY}}(R_1, a)$, which are ambiguity-independent for a fixed reference prescription.

B Extrinsic Curvature of $\chi = \text{const}$ Spheres on S^3

Consider the round metric on S^3: $d\ell^2 = a^2(d\chi^2 + \sin^2\chi\, d\Omega_2^2)$. For the 2-sphere ∂D at $\chi = \chi_0$, the outward unit normal within Σ is

$$n^i = \frac{1}{a}(\partial/\partial\chi)^i. \tag{52}$$

The induced metric on ∂D is $\sigma_{AB} = a^2 \sin^2\chi_0\, \Omega_{AB}$, where Ω_{AB} is the round metric on S^2. A direct computation of $\nabla_{(A} n_{B)}$ using the Levi–Civita connection of Σ gives

$$k = \sigma^{AB}\nabla_A n_B = \frac{2}{a}\cot\chi_0 = \frac{2}{R}\sqrt{1 - \frac{R^2}{a^2}}, \tag{53}$$

where $R = a\sin\chi_0$ is the areal radius. The Euclidean reference $k_0 = 2/R$ follows from embedding the same S^2 of radius R isometrically into $\mathbb{R}^3$.

C Zeckendorf/Fibonacci Refinement: Statement and Scope

Theorem C.1 (Zeckendorf–Lekkerkerker [40, 41])**.** *Every positive integer N has a unique representation*

$$N = \sum_j F_{k_j}, \tag{54}$$

as a sum of non-consecutive Fibonacci numbers. This representation is called the Zeckendorf representation.

The ratio $F_{n+1}/F_n \to \varphi$ (Lemma 7.1) implies that the Fibonacci ladder has φ as its unique asymptotic multiplier. A weak converse [42] establishes that uniqueness of representation essentially characterises the Fibonacci basis among monotone integer sequences.

The physical postulate in the present paper is that this refinement ladder serves as the discrete index of cosmological scale structure, via the embedding (39). This is a modelling assumption, not a theorem. The theorem-level content is the uniqueness and path-independence of the Zeckendorf decomposition. Whether this indexing is physically realised is an empirical question, tested through the predictions of Section 8.

D Observational Pipeline Details

Parameter table and priors. The minimal parameter set for a future likelihood analysis consists of:

1. Standard ΛCDM parameters: $(\Omega_b h^2, \Omega_c h^2, H_0, n_s, A_s, \tau)$ with standard flat priors.

2. Topology parameters: curvature radius R_K (or equivalently $\Omega_k < 0$) with prior $R_K \in [R_{\rm LSS}, 10^4\, c/H_0]$, and group label Γ from the spherical space-form classification.

3. (Optional) φ-embedding parameters: initial normalisation of the discrete scale ladder.

CMB pipeline. Implement closed-geometry transfer functions in CLASS [32] or CAMB [33]. Replace the standard continuous k-integral by the discrete mode sum (43). Apply matched-circle constraints for non-trivial Γ using existing code implementations [26, 27]. Couple to MCMC or nested-sampling via Cobaya [34] or MontePython [35], using official Planck likelihoods. Analyse posteriors with GetDist [36].

SGWB pipeline. Compute $(k_n^{(T)}, f_n)$ from the tensor eigenvalue relation (45) for chosen R_K. Broaden the schematic δ-comb (46) using a realistic window function accounting for finite observation time and redshift averaging. Compare to PTA [38, 39] and LVK [37] bounds via their published likelihoods or sensitivity curves.

E Model-Dependent Overlays

This appendix records model-dependent structures that can be layered on top of the GR framework but are logically independent of the core derivations in Sections 3–5. These elements should be regarded as illustrative prototypes rather than predictions of the main framework.

φ-discretised boundary scan. Define a family of areal radii $R_n = R_0 \varphi^n$ with $n \in \mathbb{Z}$, restricted to $0 \le R_n \le a(t)$ on $\Sigma \simeq S^3$. Evaluating the quasi-local energies (28) and (31) along this discrete family provides a structured quasi-local bookkeeping. This is a computational device, not a dynamical claim.

Spectral overlay. A model-dependent energy ladder $E_n^{(\zeta)} = \hbar c/(\ell_P \varphi^n)$ may be defined for phenomenological studies. Whether such a ladder has observational relevance depends on the specific matter model and lies outside the scope of the present framework.

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